



UNIVERSITÀ DI PISA

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1 DATASET

1.1 SUBSET DEL DATASET

Il dataset come da richiesta è stato preso dal primo sottotask (sub-task A) di *EVALITA 2026*, ha il formato di una tabella csv composta da una colonna che contiene una serie di testi e la seconda contenente il relativo label di classificazione binaria che identifica se il testo contenuto sia stato generato da un umano (0) o da una macchina (1). Sul sito del dataset erano presenti tre file scaricabili. Il primo file di training composto da una tabella csv con più di 33 mila entry, da questa sono stati richiesti due subset uno di training da 2000 righe e uno di validation da 1000, entrambi bilanciati, quindi con egual numero di label 0 e 1. Gli altri due file erano di test, uno contenente solo la colonna dei testi da classificare e l'altro contenente le soluzioni corrette del test set. Dal test set con label è stato richiesto di estrarre 1000 item anch'essi bilanciati tra label 0 e 1. L'estrazione bilanciata è avvenuta con l'ausilio di due contatori uno per ciascuna label di classificazione, come si può vedere dalla sezione di codice sotto riportata. Per ottenere il bilanciamento tra classi la logica di selezione delle entry prevede che i contatori abbiano lo stesso valore al termine dell'iterazione. A partire dal primo elemento della tabella csv, se il valore dei contatori è uguale viene selezionata l'entry corrente per far parte del subset e incrementato il contatore corrispondente alla label nella seconda colonna, altrimenti dall'elemento corrente si scorrono le successive entry fino a trovare la prima che abbia label corrispondente al counter di valore minore.

```
1 #[...]
2     for riga in reader:
3         #riga contiene il ['test'] e la ['label']
4         if c0+c1 < row:
5             if c0==c1:
6                 match riga["label"]:
7                     case "1":
8                         c1+=1
9                     case "0":
10                        c0+=1
11                subset.append(riga)
12                continue
13            elif c0>c1 and riga["label"] == "1":
14                c1+=1
15                subset.append(riga)
16                continue
17            elif c0<c1 and riga["label"] == "0":
```

```

18             c0+=1
19             subset.append(riga)
20             continue
21         else:
22             break
23     #[...]

```

Per l'estrazione del validation set la logica è la stessa, ma per non avere duplicati il file è stato letto al contrario, cioè a partire dall'ultima entry, infatti dato che ci sono più di 33mila campioni è improbabile che con la selezione del subset di training si arrivi in fondo alla tabella. Nella sezione di codice sotto mostro come è stata invertita la tabella.

```

1     #[...]
2     for riga in reversed(righe):
3     #[...]

```

Tuttavia trattandosi di un evento altamente improbabile ma non impossibile (non ho letto il dataset), volevo essere sicura non ci fossero elementi in comune, perciò ho performato ulteriore verifica che il training set e il validation set non avessero righe in comune. Il check è stato eseguito direttamente da terminale con un piccolo script, riportato di seguito, il quale ha confermato l'assenza di sovrapposizioni.

```

1     import pandas as pd
2
3     df1 = pd.read_csv('train_subset.csv')
4     df2 = pd.read_csv('validation_subset.csv')
5
6     # L'inner merge tiene solo ciò che è presente in ENTRAMBI i dataframe
7     if pd.merge(df1,df2, how='inner'):
8         print("ci sono elementi in comune")
9     else:
10        print("non ci sono elemnti in comune")

```

1.2 FEATURES

1.2.1 N-GRAMMI

sleek is the main package. It imports the packages (see Table A.1) and setups the settings that make Sleek Template easy to use. The package provides two options:

1. header adds a header with the current chapter to pages;
2. french makes the French setup consistent with other languages.

However, nothing prevents you from tweaking the settings to your liking in the source code. For example, you can change the font of the document from “Linux Libertine” to “Latin Modern Roman”, by replacing

```
% Font
\RequirePackage{libertine}
\RequirePackage{libertinustlmath}
\RequirePackage[T1]{fontenc}
```

with

```
% Font
\RequirePackage{lmodern}
\RequirePackage[T1]{fontenc}
```

in the `sleek.sty` file.

1.2.2 MATHEMATICS

This template uses `amsmath` and `amssymb`, which are the de-facto standard for typesetting mathematics. Additionally, `esint` provides alternative integral symbols (see Table 78 in [1]) and `bm` is used for bold math symbols like vectors (see (1.1)).

A few custom macros have also been added such as `\rbk`, `\sbk` and `\cbk` for respectively round, square and curly brackets or `\abs` for absolute value. Here follows some examples showcasing what is possible with the default packages of `sleek`.

$$(\pi), \quad \left\lceil \frac{\pi}{2} \right\rceil, \quad \left\{ \pi = 3 + \frac{1}{7 + \frac{1}{15 + \frac{1}{1 + \dots}}} \right\}, \quad |2\pi|, \quad \left\| \frac{\pi}{4} \right\|, \quad \left\lceil \frac{22}{7} \right\rceil, \quad \left\lfloor \frac{\pi^2}{6} \right\rfloor$$

$$\oiint_S \mathbf{E} \cdot d\mathbf{s} = \iiint_V \frac{\rho}{\epsilon_0} dV \quad (1.1)$$

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

$$\frac{dx}{dt} = \alpha x - \beta xy \quad (1.2a)$$

$$\frac{dy}{dt} = \delta xy - \gamma y \quad (1.2b)$$

$$\ln|x| + C = \int \frac{1}{x} dx$$

$$\exp(x) = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n$$

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases} \quad (1.3)$$

$$\begin{aligned} P(A, B) &= P(A | B)P(B) \\ \Leftrightarrow P(A | B) &= \frac{P(A, B)}{P(B)} \\ &= P(B | A) \frac{P(A)}{P(B)} \end{aligned}$$

1.2.3 UNITS

The `siunitx` package [2] provides three commands to typeset numbers, units and quantities: `\num`, `\unit` and `\qty`.

It is possible to write, both in text or math modes, numbers without units (e.g. 1, 1.0, −1, 3.141 59, 10^{100} , $N_A = 6.022 \times 10^{23}$), units without quantity (e.g. J = N m = kg m² s^{−2}) and, finally, quantities with their units (e.g. 9.81 m s^{−2}, $c = 299.6 \times 10^6$ m s^{−1}).

1.2.4 LISTS

Sleek Template uses `enumitem` to enhance the listing capabilities of $\LaTeX$. There are several lists environments:

- `itemize` for unordered lists;
- `enumerate` for ordered lists;
- `description` for descriptive lists.

In a list, each element is preceded by the command `\item`. It is possible to modify the labels 1. individually with `\item[newLabel]` or 2. for the whole environment with the `label=newLabel` option.

In the case of `enumerate`, `newLabel` can contain special expressions (see Table A.4) that will adapt to the item number. For example, `label=(\alph*)` defines the label sequence “(a), (b), (c), ...”. Still in the case of `enumerate`, the `\setcounter` and `\addtocounter` commands allow to modify the current item number.

One could want to reduce the space between items with the `noitemsep` option or to delete the left margin with the `leftmargin=*` option.

It is also possible to write nested lists. Here follows a very condensed example.

- Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Arcu ac tortor dignissim convallis aenean et tortor. In eu mi bibendum neque egestas congue quisque.

- + Semper quis lectus nulla at volutpat diam ut. Felis eget velit aliquet sagittis id. Blandit aliquam etiam erat velit scelerisque in dictum non consectetur.

$$a^2 + b^2 = c^2 \quad (1.4)$$

- Nibh sed pulvinar proin gravida hendrerit lectus. Pretium aenean pharetra magna ac placerat vestibulum lectus mauris. Non consectetur a erat nam at lectus urna duis.
 - i. Nibh tortor id aliquet lectus. Sit amet justo donec enim diam vulputate ut pharetra sit.
 - iv. Condimentum id venenatis a condimentum vitae. Quis eleifend quam adipiscing vitae proin sagittis nisl.
 - xx. Proin sagittis nisl rhoncus mattis rhoncus urna neque viverra.

- Elit scelerisque mauris pellentesque pulvinar pellentesque habitant morbi tristique senectus.

Ridiculus mus mauris vitae ultricies leo. Mollis aliquam ut porttitor leo a diam. Velit egestas dui id ornare arcu odio ut sem nulla.

Nullam vehicula ipsum a arcu. Nibh sit amet commodo nulla facilisi nullam. At erat pellentesque adipiscing commodo elit. Libero volutpat sed cras ornare arcu dui.

1.2.5 FIGURES

Thanks to the `graphicx` package, it is possible to include external graphic documents (images, plots, etc.) in your document with the `\includegraphics` command. Most image type format (jpg, png, bmp, etc.) are supported by this command. However, it should be noted that it is highly preferable to use vectorial types, such as pdf or eps.



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Figure 1.1. Random University logo.

1.2.6 TABLES

The `multirow` package comes handy for complex table formatting such as multi-column or multi-row cells.

a			qrs
b	ef	jkl	tuvx
cd	ghi	mnp	wyz

Table 1.1. Example of multi-column cells.

a	b	c
	de	fg
	hij	klm
nopq	rstuv	wxyz

Table 1.2. Example of multi-row cells.

The `booktabs` package also provides tools to build publication-ready tables. The tables in Appendix [A](#) are good examples.

1.3 sleek-title

Sleek Template offers a custom title-page with the package `sleek-title`. The formatting of the title-page is automatically inferred from the fields that the user has provided.

The fields are `\logo`, `\institute`, `\faculty`, `\department`, `\title`, `\subtitle`, `\author`, `\supervisor`, `\context` and `\date`.

Among these, only `\title`, `\author` and `\date` have to be provided. However, none of the fields should stay empty. Prefer deleting or commenting the line if so.

```
\logo{./resources/pdf/logo.pdf}
\institute{Random University}
\faculty{Faculty of Whatever Sciences}
%\department{Department of Anything but Psychology}
\title{A sleek \LaTeX{} template}
\subtitle{With a sleeker title-page}
\author{\textit{Author}\\Francois \textsc{Rozet}}
%\supervisor{Linus \textsc{Torvalds}}
%\context{A long time ago in a galaxy far, far away...}
\date{\today}
```

Listing 1.1. Example of `sleek-title` title-page definition.

It is also possible to use Sleek Template without `sleek-title`, in which case the default $\text{\LaTeX}$ title-page will be used.

1.4 sleek-theorems

sleek-theorems is based on the amsthm and thmtools packages. It provides a handful of theorem-like environments, each of which have different style and purpose: theorem, lemma, proposition, proof, definition, hypothesis, method, question, answer, example, remark, note and tip.

Note. The french and german options translate the names of the provided environments. It is also possible, and easy, to add your own language as an option in the source code.

Theorem 1 (Triangle inequality). *Let be a triangle in Euclidean space. Then the sum of the lengths of two of its sides always surpass or equals the length of the third.*

Proof 1.1. Let a , b and c be the lengths of the sides of a triangle in Euclidean space and α , β , γ their respective opposite angle. By the generalized Pythagoras' theorem, we have

$$\begin{aligned} c^2 &= a^2 + b^2 - 2ab \cos \gamma \\ &\leq a^2 + b^2 + 2ab \\ &\leq (a + b)^2 \\ \Leftrightarrow \quad c &\leq a + b \end{aligned}$$

Therefore in any triangle, the sum of the lengths of two sides always surpass or equals the length of the third. ■

In addition, these environments also have framed versions – fmd-theorem, fmd-lemma, etc. – for better encapsulation.

Theorem 2. Triangle inequality

Let be a triangle in Euclidean space. Then the sum of the lengths of two of its sides always surpass or equals the length of the third.

Proof 2.1

Let a , b and c be the lengths of the sides of a triangle in Euclidean space and α , β , γ their respective opposite angle. By the generalized Pythagoras' theorem, we have

$$\begin{aligned} c^2 &= a^2 + b^2 - 2ab \cos \gamma \\ &\leq a^2 + b^2 + 2ab \\ &\leq (a + b)^2 \\ \Leftrightarrow \quad c &\leq a + b \end{aligned}$$

Therefore in any triangle, the sum of the lengths of two sides always surpass or equals the length of the third. ■

Question

Based on the theorem 2, what is the shortest path from a point A to a point B in Euclidean geometry ?

1.5 sleek-listings

The `sleek-listings` package is a small collection of styles and languages for the `listings` package, which is useful to showcase nicely samples of code. The package provides the styles `default`, `numbered`, `framed`, `ruled`, `framednumbered` and `rulednumbered` and the languages `latex`, `c*`, `cpp*`, `matlab*`, `python*`, `java*` and `scheme`, with a simple color scheme for keywords and comments.

For instance, if you want to showcase Python code within a ruled code block (top and bottom rule) with line numbers, you can either import a `.py` file with

```
\lstinputlisting[style=rulednumbered, language=python*]{filename.py}
```

or paste the code within the $\LaTeX$ document as

```
1 import numpy as np # Unnecessary import
2
3 a, b = 69., .420
4
5 def f(a: float, b: float) -> float:
6     r"""Sum two numbers
7
8     Parameters
9     -----
10    a: first number
11    b: second number
12
13    Returns
14    -----
15    the sum of 'a' and 'b'
16    """
17
18    return a + b
19
20 c = f(a, b)
21
22 print('{:f} + {:f} equals {:f}'.format(a, b, c))
```

BIBLIOGRAPHY

- [1] Scott Pakin. “The comprehensive L^AT_EX symbol list”. 2020. URL: <http://tug.ctan.org/info/symbols/comprehensive/symbols-a4.pdf>.
- [2] Joseph Wright. “siunitx – A comprehensive (SI) units package”. 2022. URL: <https://ctan.org/pkg/siunitx>.

A TABLES

Package	Purpose
amsmath	Mathematical typesetting
amsthm	Mathematical environments for theorems, proofs, etc.
booktabs	Weighted rules for tables
biblatex	Bibliography
csquotes	Inline and display quotations
enumitem	Lists and enumerations
float	Floating objects such as figures and tables
graphicx	Graphics
hyperref	Hyperlinks and bookmarks
listings	Code listings
multirow	Table cells that span multiple columns or rows
siunitx	Typesetting of units
subcaption	Sub-figures and sub-captions

Table A.1. List of the most relevant packages imported by Sleek Template.

<code>{\rmfamily abcABC123}</code>	abcABC123
<code>{\sffamily abcABC123}</code>	abcABC123
<code>{\ttfamily abcABC123}</code>	abcABC123
<code>{\mdseries abcABC123}</code>	abcABC123
<code>{\bfseries abcABC123}</code>	abcABC123
<code>{\itshape abcABC123}</code>	<i>abcABC123</i>
<code>{\scshape abcABC123}</code>	ABCABC123
<code>{\slshape abcABC123}</code>	<i>abcABC123</i>
<code>{\upshape abcABC123}</code>	abcABC123
<code>\emph{abcABC123}</code>	<i>abcABC123</i>
<code>\lowercase{abcABC123}</code>	abcabc123
<code>\normalfont{abcABC123}</code>	abcABC123
<code>\textrm{abcABC123}</code>	abcABC123
<code>\textbf{abcABC123}</code>	abcABC123
<code>\textit{abcABC123}</code>	<i>abcABC123</i>
<code>\textsc{abcABC123}</code>	ABCABC123
<code>\textsf{abcABC123}</code>	abcABC123
<code>\textsl{abcABC123}</code>	<i>abcABC123</i>
<code>\textsubscript{abcABC123}</code>	abcABC123
<code>abcABC123</code>	abcABC123
<code>\texttt{abcABC123}</code>	abcABC123
<code>\underline{abcABC123}</code>	<u>abcABC123</u>
<code>\uppercase{abcABC123}</code>	ABCABC123

Table A.2. Available text fonts in $\text{\LaTeX}$.

<code>\$_{\mathcal{abcABC123}}\$</code>	$\mathcal{ABC}$
<code>\$_{\mathit{abcABC123}}\$</code>	$\mathit{abcABC123}$
<code>\$_{\mathnormal{abcABC123}}\$</code>	$\mathnormal{abcABC123}$
<code>\$_{\mathrm{abcABC123}}\$</code>	$\mathrm{abcABC123}$
<code>\$_{\mathbb{abcABC123}}\$</code>	$\mathbb{abcABC123}$
<code>\$_{\mathfrak{abcABC123}}\$</code>	$\mathfrak{abcABC123}$

Table A.3. Available math fonts in $\text{\LaTeX}$ and AMS.

Expression	Description
<code>\arabic*</code>	Arabic numbers (1, 2, 3, ...)
<code>\alph*</code>	Lowercase letters (a, b, c, ...)
<code>\Alph*</code>	Uppercase letters (A, B, C, ...)
<code>\roman*</code>	Lowercase Roman numerals (i, ii, iii, ...)
<code>\Roman*</code>	Uppercase Roman numerals (I, II, III, ...)

Table A.4. Special expressions for the label of enumerate environments.